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An isotopic error compensation method for rovibrational spectra of isotopic diatomic molecules

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An isotopic error compensation (IEC) approach is presented to study the accurate rovibrational spectra for isotopic diatomic molecules based on the theoretical analysis of the errors of the rovibrational term values $E^{\mathrm{cal}}(\upsilon, J)$ from the same effective internuclear potential. The IEC term value is defined by $E_{\beta}^{\mathrm{IEC}}(\upsilon, J) \equiv E_{\beta}^{\mathrm{cal}}(\upsilon, J) + [E_{\alpha}^{\mathrm{exp}}(\upsilon, J) - E_{\alpha}^{\mathrm{cal}}(\upsilon, J)]$, where α and β indicate isotopic molecules with reduced masses μ_{α} and μ_{β} respectively. This approach is of special use for $\mu_{\alpha} \approx \mu_{\beta}$. The IEC approach is also an effective way to eliminate large systematic errors introduced from the effective potential and the errors introduced by some Born–Oppenheimer breakdown effects. The results show that the IEC approach can be used to predict more accurate spectra for isotopic molecules than the semi-classical isotope relation does. The predicted IEC spectra are influenced by the accuracies of the observed isotopic molecule spectra, as well as the effective potential for the isotopic molecules used to calculate the IEC spectra. Some new IEC spectra data are predicted for the (1,0)–(4,3) bands of ^{6}LiH and the (1,0)–(5,4) bands of ^{6}LiD ground state, the uncertainties for most of them are estimated to be 0.0015–0.0020 cm^{-1}.

Keywords: Isotope effect; Diatomic molecule; Molecular spectra; LiH

1. Introduction

The rovibrational energies $E(\upsilon, J)$ of a $^{1}\Sigma$ electronic state of a diatomic molecule BA can usually be represented by the Dunham power series [1]

$$E(\upsilon, J) = \sum\nolimits_{l,m} Y_{l,m}\left(\upsilon + \frac{1}{2}\right)^{l} [J(J+1)]^{m} \qquad (1)$$

where υ and J are the vibrational and the rotational quantum numbers respectively. The Dunham coefficients $Y_{l,m}$ can be determined using known experimental spectral data and a numerical fitting approach. For related isotopic molecules the mass-dependent Dunham coefficients $Y_{l,m}$ and the mass-independent Dunham coefficients $U_{l,m}$ are linked semi-classically by (where μ is the reduced mass of the molecule and $U_{l,m}$ is isotopically invariant) [1–3]

$$Y_{l,m} = \mu^{-(l+2m)/2} U_{l,m} \qquad (2)$$

subject to two approximations: (i) the related isotopes experience exactly the same effective internuclear potential function, and (ii) the first-order WKB approximation is exact for the studied isotopes. A modified form of equation (2) including some higher order WKB correction terms was also given by Dunham [1]. However, both equation (2) and its modified form given by Dunham are not adequate for reliable high resolution spectra. These spectra require new correction terms involving Born–Oppenheimer (BO) breakdown [4] in equation (2). Such correction terms had been involved in the general multi-isotopomer Dunham-type energy expression, which was originally expressed by Ross *et al*. [5]:

$$E(\upsilon, J) = \sum\nolimits_{l,m} \mu^{-(1+2m)/2} \times \left[1 + \left(\frac{m_e}{M_A}\right)\Delta_{l,m}^{A} + \left(\frac{m_e}{M_B}\right)\Delta_{l,m}^{B}\right] U_{l,m} \times \left(\upsilon + \frac{1}{2}\right)^{l} [J(J+1)]^{m} \qquad (3)$$

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for which Bunker [6] and Watson [2] later provided theoretical justification. For related isotopic diatomic molecules, according to equation (3), the $Y_{l,m}$ are given by [2]

$$Y_{l,m} = \mu^{-(l+2m)/2}\left[1 + \left(\frac{m_e}{M_A}\right)\cdot\Delta^A_{l,m} + \left(\frac{m_e}{M_B}\right)\cdot\Delta^B_{l,m}\right]\cdot U_{l,m} \tag{4}$$

where m_e is the electron mass, M_A and M_B are the A and B atomic masses, the $\Delta^i_{l,m}(i = A, B)$ are mass-independent Ross–Watson parameters which are isotopic invariant and are used to correct the BO breakdown on the A and B atom centres. [2] As an effective representation of rovibrational energies, equation (3) is widely applied to fit observed spectra [7–11].

An alternative numerical approach known as the "direct potential fits" (DPF) method developed by Coxon and Hajigeorgiou [12] has been commonly used to analyse transition frequencies of isotopic diatomic molecules [13–18]. In this approach the Schrödinger equation is solved for a Hamiltonian model including a parameterized potential energy function and BO breakdown functions. The differences between its eigenvalues are compared with the observed frequencies to optimize the parameters in the model. The corresponding Hamiltonian model of equation (3) introduced by Watson [2] is widely used in the DPF approach and has achieved remarkable successes. Although the procedure is rather complicated, the DPF method has a better ability to predict transition frequencies outside the range analysed. On the other hand, it is known that the potential parameters, especially the BO breakdown functions, depend on the observed spectroscopic line positions for isotopes of the same species.

Although equation (3) works as a good representation of rovibrational energies $E(\upsilon,J)$ for isotopes, only a few of the coefficients $\Delta_{l,m}$ with low values of l and m can be determined from experimental data. As pointed out by Le Roy [19], fitting cannot yield $U_{l,m}$ truly independent of mass unless data are available for isotopomers formed from more than one isotope of both of the atoms forming the studied molecule. As a result, although formally mass independent, fitted $U_{l,m}$ parameters actually depend on the isotopic range of the data available for analysis. And there is no direct relation between $U_{l,m}$ and the Ross–Watson parameters $\Delta^i_{l,m}$ [2, 19, 20]. While ignoring the correction terms in equations (3) and (4), one may not obtain more accurate results than those from the semi-classical isotope relation (SCIR), equation (2). Then on condition that the spectral data are available for only one isotopomer, how could one predict accurate spectral data for other isotopomers? The same problem is also associated with Le Roy's [19] modified form of equation (3). To address this problem is the motivation of the present work.

An isotopic error compensation (IEC) method is suggested based on analysis of the errors of the $E(\upsilon, J)$ expressed in equation (3) to predict accurate isotopic molecular spectra. The new data of the molecule studied are produced from the effective internuclear potential and accurate spectral data of its isotopic molecule. A theoretical analysis of the IEC approach is presented in section 2, and the results of its application to the ground states of ^{6,7}Li^{1,2}H isotopomers are presented and discussed in section 3. Section 4 summarizes the present work.

2. Isotopic error compensation (IEC) approach

The effective radial Schrödinger equation (3) for the $^1\Sigma$ electronic ground state of isotopic molecules B_iA can be written in the following form [8]

$$\begin{aligned}&\left\{-\frac{\hbar^2}{2\mu_i}\frac{d^2}{\mathrm{d}R^2} + V_i^{\mathrm{eff}}(R) + \frac{\hbar^2}{2\mu_i}\frac{J(J+1)}{R^2}\right.\\ &\left.\left[1 + \frac{m_e}{M_A}q^A(R) + \frac{m_e}{M_{B_i}}q^B(R)\right]\right\}\\ &\Psi_i(R;\upsilon,J) = E_i(\upsilon,J)\Psi_i(R;\upsilon,J)\end{aligned} \tag{5}$$

where $i = \alpha$, β indicates different isotopic molecules or isotopes, $V_i^{\mathrm{eff}}(R)$ is the effective radial potential with the form

$$V_i^{\mathrm{eff}}(R) = V_{\mathrm{BO}}(R) + \frac{m_e}{M_A}U^A(R) + \frac{m_e}{M_{B_i}}U^B(R) \tag{6}$$

$V_{\mathrm{BO}}(R)$ is the BO approximate potential, $U^A(R)$, $U^B(R)$, $q^A(R)$ and $q^B(R)$ are isotopically invariant. $U^A(R)$ and $U^B(R)$ are the J-independent, and $q^A(R)$ and $q^B(R)$ are the J-dependent BO breakdown correction functions on the A and B atom centres respectively. [8]

For the isotopic molecule $B_\alpha A$, the accurate eigenvalues of equation (5) can be taken as the accurate experimental rovibrational term values, which can be effectively expanded as the following series according to equation (3):

$$\begin{aligned}E_\alpha(\upsilon,J) = &\sum\nolimits_{l,m}\mu_\alpha^{-(l+2m)/2}\\ &\times\left[1 + \left(\frac{m_e}{M_A}\right)\cdot\Delta^A_{l,m} + \left(\frac{m_e}{M_{B\alpha}}\right)\cdot\Delta^B_{l,m}\right]\\ &\cdot\ U_{l,m}\{\upsilon^l J^m\}\end{aligned} \tag{7}$$

where $\{\upsilon^l J^m\} = (\upsilon + 1/2)^l[J(J+1)]^m$.

Similarly, for the isotopic molecule $B_\beta A$, the corresponding eigenvalues of equation (5) and the accurate experimental rovibrational term values can be expressed as

$$E_\beta(\upsilon, J) = \sum\nolimits_{l,m} \mu_\beta^{-(l+2m)/2} \times \left[1 + \left(\frac{m_e}{M_A}\right) \cdot \Delta_{l,m}^A + \left(\frac{m_e}{M_{B_\beta}}\right) \cdot \Delta_{l,m}^B\right] \cdot U_{l,m}\{\upsilon^l J^m\}. \quad (8)$$

The IEC approach is stated based on the following discussion. Using the accurate eigenvalue expression for the $B_\alpha A$ molecule in equation (7) and the SCIR in equation (2), the semi-classical approximate rovibrational term values for the $B_\beta A$ molecule can be expressed as

$$E_\beta^{\mathrm{SCIR}}(\upsilon, J) = \sum\nolimits_{l,m} \mu_\beta^{-(l+2m)/2} \times \left[1 + \left(\frac{m_e}{M_A}\right) \cdot \Delta_{l,m}^A + \left(\frac{m_e}{M_{B_\alpha}}\right) \cdot \Delta_{l,m}^B\right] \cdot U_{l,m}\{\upsilon^l J^m\} \quad (9)$$

which differs from equation (8), in which M_{B_β} is replaced by M_{B_α}. As is well known, the results from equation (9) contain some errors that cannot be neglected for accurate molecular spectra, including the errors introduced from the first-order WKB approximation [1, 19].

According to equation (6), the difference between $V_\alpha^{\mathrm{eff}}(R)$ and $V_\beta^{\mathrm{eff}}(R)$ is $(m_e/M_{B_\alpha} - m_e/M_{B_\beta}) \cdot U^B(R)$, and it may be very small, therefore $V_\alpha^{\mathrm{eff}}(R)$ may be a good approximation of $V_\beta^{\mathrm{eff}}(R)$. Similarly, for $M_{B_\beta} \approx M_{B_\alpha}$ the J-dependent BO correction for B_α, $q^B(R)/M_{B\alpha}$, is also a good approximation for the J-dependent BO correction of B_β, $q^B(R)/M_{B\beta}$. For $B_\beta A$, if the effective rotationless potential $V_\beta^{\mathrm{eff}}(R)$ and the J-dependent BO correction are both substituted by those of $B_\alpha A$, the corresponding Hamiltonian can be written as

$$\hat{H}'_\beta = -\frac{\hbar^2}{2\mu_\beta}\frac{d^2}{\mathrm{d}R^2} + \left[V_{\mathrm{BO}}(R) + \frac{m_e}{M_A} U^A(R) + \frac{m_e}{M_{B_\alpha}} U^B(R)\right] + \frac{\hbar^2}{2\mu_\beta}\frac{J(J+1)}{R^2}\left[1 + \frac{m_e}{M_A} q^A(R) + \frac{m_e}{M_{B_\alpha}} q^B(R)\right]. \quad (10)$$

The corresponding accurate eigenvalues of the Schrödinger equation with the Hamiltonian in equation (10) can be effectively expressed as

$$E'_\beta(\upsilon, J) = \sum\nolimits_{l,m} \mu_\beta^{-(l+2m)/2} \times \left[1 + \left(\frac{m_e}{M_A}\right) \cdot \Delta_{l,m}^A + \left(\frac{m_e}{M_{B_{\beta'}}}\right) \cdot \Delta_{l,m}^B\right] \cdot U_{l,m}\{\upsilon^l J^m\}. \quad (11)$$

where $M_{B_{\beta'}}$ is a value in mass unit, it is neither M_{B_β} nor M_{B_α}, and $M_{B_{\beta'}}$ is expected to locate between M_{B_β} and M_{B_α}, i.e.

$$M_{B_\beta} < M_{B_{\beta'}} < M_{B_\alpha} \quad \text{or} \quad M_{B_\beta} > M_{B_{\beta'}} > M_{B_\alpha}. \quad (12)$$

Equation (11) corresponds to the same molecular potential as that for equation (7), but it does not contain the errors introduced from the first-order WKB approximation. Compared with the exact expression of equation (8) for the $B_\beta A$ molecule, equation (11) presents more accurate results than the SCIR results, equation (9).

When accurate rotationless potential $V_\alpha^{\mathrm{eff}}(R)$ is not available, one can use a trial potential $U_\alpha^{\mathrm{eff}}(R)$,

$$U_\alpha^{\mathrm{eff}}(R) = U_{\mathrm{BO}}(R) + \frac{m_e}{M_A} U^A(R) + \frac{m_e}{M_{B_\alpha}} U^B(R) \quad (13)$$

where $U_{\mathrm{BO}}(R)$ is the trial BO potential. $V_\alpha^{\mathrm{eff}}(R)$ and $U_\alpha^{\mathrm{eff}}(R)$ have the same BO breakdown correction functions, but they may have different BO potentials. This can be easily manipulated since both $V_\alpha^{\mathrm{eff}}(R)$ and $U_\alpha^{\mathrm{eff}}(R)$ are functions of the single variable R. For simplicity the J-dependent BO breakdown correction functions $q^{A,B}(R)$ can be omitted for both isotopic molecules $B_\alpha A$ and $B_\beta A$ from their corresponding Hamiltonians, but their eigenvalues of equation (5) are also expressed as the forms of equation (7) and equation (11), respectively. For $J = 0$ it gives the same results because the J-dependent functions $q^{A,B}(R)$ are considered in the effective Hamiltonian of the Schrödinger equation equation (5). The calculated results for the $B_\alpha A$ molecule from a good trial effective potential $U_\alpha^{\mathrm{eff}}(R)$ can then be expanded as

$$E_\alpha^{\mathrm{cal}}(\upsilon, J) = \sum\nolimits_{l,m} \mu_\alpha^{-(l+2m)/2} \times \left[1 + \left(\frac{m_e}{M_A}\right) \cdot \Delta_{l,m}^A + \left(\frac{m_e}{M_{B_\alpha}}\right) \cdot \Delta_{l,m}^B\right] \cdot U_{l,m}^{\mathrm{cal}} \cdot \{\upsilon^l J^m\}. \quad (14)$$

In equation (14) all the errors caused by the $U_{\mathrm{BO}}(R)$ are included in the $U_{l,m}^{\mathrm{cal}}$ mathematically. Like equation (11), the calculated rovibrational energy for $B_\beta A$ from the *same* trial potential $U_\alpha^{\mathrm{eff}}(R)$ can be expanded as

$$E_\beta^{\mathrm{cal}}(\upsilon,J)=\sum\nolimits_{l,m}\mu_\beta^{-(l+2m)/2}\times\left[1+\left(\frac{m_e}{M_A}\right)\cdot\Delta_{l,m}^{A}+\left(\frac{m_e}{M_{B_{\beta'}}}\right)\cdot\Delta_{l,m}^{B}\right]\cdot U_{l,m}^{\mathrm{cal}}\cdot\left\{\upsilon^l J^m\right\} \tag{15}$$

where $M_{B'_\beta}$ is indicated as in equation (12). Using equations (7) and (8), the errors of $E_\alpha^{\mathrm{cal}}(\upsilon,J)$ and $E_\beta^{\mathrm{cal}}(\upsilon,J)$ can then be expressed as

$$\begin{aligned}\delta E_\alpha^{\mathrm{cal}}(\upsilon,J)&\equiv E_\alpha(\upsilon,J)-E_\alpha^{\mathrm{cal}}(\upsilon,J)\\&=\sum\nolimits_{l,m}\mu_\alpha^{-(l+2m)/2}\times\left[1+\left(\frac{m_e}{M_A}\right)\Delta_{l,m}^{A}+\left(\frac{m_e}{M_{B_\alpha}}\right)\Delta_{l,m}^{B}\right]\times\left[U_{l,m}-U_{l,m}^{\mathrm{cal}}\right]\left\{\upsilon^l J^m\right\}\end{aligned} \tag{16}$$

$$\begin{aligned}\delta E_\beta^{\mathrm{cal}}(\upsilon,J)&\equiv E_\beta(\upsilon,J)-E_\beta^{\mathrm{cal}}(\upsilon,J)\\&=\sum\nolimits_{l,m}\mu_\beta^{-(l+2m)/2}\times\left[1+\left(\frac{m_e}{M_A}\right)\cdot\Delta_{l,m}^{A}+\left(\frac{m_e}{M_{B_\beta}}\right)\cdot\Delta_{l,m}^{B}\right]\cdot\left[U_{l,m}-U_{l,m}^{\mathrm{cal}}\right]\cdot\left\{\upsilon^l J^m\right\}\\&\quad+\sum\nolimits_{l,m}\mu_\beta^{-(l+2m)/2}\left[\left(\frac{m_e}{M_{B_\beta}}\right)-\left(\frac{m_e}{M_{B_{\beta'}}}\right)\right]\cdot\Delta_{l,m}^{B}\cdot U_{l,m}^{\mathrm{cal}}\cdot\left\{\upsilon^l J^m\right\}.\end{aligned} \tag{17}$$

On condition that $M_{B_\beta}\approx M_{B_\alpha}$ (therefore $\mu_\alpha\approx\mu_\beta$ and $M_{B_\beta}\approx M_{B'_\beta}$, we have

$$\delta E_\alpha^{\mathrm{cal}}(\upsilon,J)\approx\delta E_\beta^{\mathrm{cal}}(\upsilon,J) \tag{18}$$

or

$$\delta E_\beta^{\mathrm{cal}}(\upsilon,J)-\delta E_\alpha^{\mathrm{cal}}(\upsilon,J)\rightarrow\text{small values} \tag{19}$$

The interesting questions are: how small are the values of equation (19)? and can equation (19) give smaller values than the errors of the semi-classical isotope effect results $E_\beta^{\mathrm{SCIR}}(\upsilon,J)$ in equation (9)? The following will demonstrate positive answers. Using equations (8) and (9), the error in $E_\beta^{\mathrm{SCIR}}(\upsilon,J)$ is

$$\begin{aligned}\delta E_\beta^{\mathrm{SCIR}}(\upsilon,J)&=E_\beta(\upsilon,J)-E_\beta^{\mathrm{SCIR}}(\upsilon,J)\\&=\sum\nolimits_{l,m}\mu_\beta^{-(l+2m)/2}\times\left[\left(\frac{m_e}{M_{B_\beta}}\right)-\left(\frac{m_e}{M_{B_\alpha}}\right)\right]\cdot\Delta_{l,m}^{B}\cdot U_{l,m}\cdot\left\{\upsilon^l J^m\right\}.\end{aligned} \tag{20}$$

Using equation (16) and equation (17), the value of equation (19) is

$$\begin{aligned}&\delta E_\beta^{\mathrm{cal}}(\upsilon,J)-\delta E_\alpha^{\mathrm{cal}}(\upsilon,J)\\&=\sum\nolimits_{l,m}\left[\mu_\beta^{-(l+2m)/2}-\mu_\alpha^{-(l+2m)/2}\right]\cdot\left[1+\left(\frac{m_e}{M_A}\right)\cdot\Delta_{l,m}^{A}\right]\cdot\left[U_{l,m}-U_{l,m}^{\mathrm{cal}}\right]\left\{\upsilon^l J^m\right\}\\&\quad+\sum\nolimits_{l,m}\left[\mu_\beta^{-(l+2m)/2}\left(\frac{m_e}{M_{B_\beta}}\right)-\mu_\alpha^{-(l+2m)/2}\left(\frac{m_e}{M_{B_\alpha}}\right)\right]\cdot\Delta_{l,m}^{B}\cdot\left[U_{l,m}-U_{l,m}^{\mathrm{cal}}\right]\left\{\upsilon^l J^m\right\}\\&\quad+\sum\nolimits_{l,m}\mu_\beta^{-(l+2m)/2}\left[\left(\frac{m_e}{M_{B_\beta}}\right)-\left(\frac{m_e}{M_{B'_\beta}}\right)\right]\cdot\Delta_{l,m}^{B}\cdot U_{l,m}^{\mathrm{cal}}\left\{\upsilon^l J^m\right\}.\end{aligned} \tag{21}$$

On condition that $U_{l,m}^{\mathrm{cal}}\rightarrow U_{l,m}$, i.e. the trial effective potential $U_\alpha^{\mathrm{eff}}(R)$ is accurate enough, $U_\alpha^{\mathrm{eff}}(R)\rightarrow V_\alpha^{\mathrm{eff}}(R)$, the first two summations on the right-hand side of equation (21) can be omitted, therefore,

$$\begin{aligned}\delta E_\beta^{\mathrm{cal}}(\upsilon,J)-\delta E_\alpha^{\mathrm{cal}}(\upsilon,J)&\approx\sum\nolimits_{l,m}\mu_\beta^{-(l+2m)/2}\times\left[\left(\frac{m_e}{M_{B_\beta}}\right)-\left(\frac{m_e}{M_{B_{\beta'}}}\right)\right]\cdot\Delta_{l,m}^{B}\\&\quad\cdot U_{l,m}^{\mathrm{cal}}\left\{\upsilon^l J^m\right\}\end{aligned} \tag{22}$$

when $U_\alpha^{\mathrm{eff}}(R)=V_\alpha^{\mathrm{eff}}(R)$, then $\approx$ can be changed into $=$ in equation (22). Comparing equation (20) with equation (22), and using equation (12) we have,

$$\frac{\delta E_\beta^{\mathrm{cal}}(\upsilon,J)-\delta E_\alpha^{\mathrm{cal}}(\upsilon,J)}{\delta E_\beta^{\mathrm{SCIR}}(\upsilon,J)}\approx\frac{M_{B'_\beta}-M_{B_\beta}}{M_{B_\alpha}-M_{B_\beta}}\cdot\frac{M_{B_\alpha}}{M_{B'_\beta}}<1 \tag{23}$$

From equation (12) it is clear that the value of equation (23) should be greater than zero. As a result, equation (21) or equation (22) can give smaller errors than equation (20) does, on condition that the effective potential $U_\alpha^{\rm eff}(R)$ for the $B_\alpha A$ molecule is accurate enough.

Based on the above analysis on isotopic molecular spectra errors, the IEC rovibrational energies $E^{\rm IEC}(\upsilon,J)$ are given by

$$E_\beta^{\rm IEC}(\upsilon,J) = E_\beta(\upsilon,J) - \left[\delta E_\beta^{\rm cal}(\upsilon,J) - \delta E_\alpha^{\rm cal}(\upsilon,J)\right]. \quad (24)$$

The error of the IEC rovibrational energies $E^{\rm IEC}(\upsilon,J)$ is defined as

$$\delta E_\beta^{\rm IEC}(\upsilon,J) \equiv E_\beta(\upsilon,J) - E_\beta^{\rm IEC}(\upsilon,J). \quad (25)$$

From equation (24) it is clear that

$$\delta E_\beta^{\rm IEC}(\upsilon,J) = \delta E_\beta^{\rm cal}(\upsilon,J) - \delta E_\alpha^{\rm cal}(\upsilon,J) \quad (26)$$

Using the definitions of equation (16) and equation (17), one has

$$E_\beta^{\rm IEC}(\upsilon,J) \equiv E_\beta^{\rm cal}(\upsilon,J) + \left[E_\alpha(\upsilon,J) - E_\alpha^{\rm cal}(\upsilon,J)\right] \quad (27)$$

which indicates that the IEC rovibrational energies for molecule $B_\beta A$ can be given by the sum of the approximate theoretical energy $E_\beta^{\rm cal}(\upsilon,J)$ of $B_\beta A$ and the small energy difference $\delta E_\alpha^{\rm cal}(\upsilon,J)$ of its isotopic molecule $B_\alpha A$. Equation (27) can give more accurate energy than the semi-classical isotope energy formula equation (9), and so does the $(\upsilon',J')\rightarrow(\upsilon,J)$ transition of the corresponding IEC rovibrational spectra

$$\Delta E^{\rm IEC}(\upsilon',J';\upsilon,J) = E^{\rm IEC}(\upsilon',J') - E^{\rm IEC}(\upsilon,J). \quad (28)$$

For $J=J'-1$ and $J=J'+1$, equation (28) gives the R-branch and the P-branch spectrum respectively for $(\upsilon',\ \upsilon)$ transition band. When accurate experimental spectra of $B_\alpha A$ are available, the P-branch spectrum of $B_\beta A$ can be expressed using equation (27) as follows:

$$P^{\rm IEC}(\upsilon',\upsilon;J;B_\beta A) = P^{\rm cal}(\upsilon',\upsilon;J;B_\beta A) + \delta P^{\rm cal}(\upsilon',\upsilon;J;B_\alpha A) \quad (29)$$

where

$$\delta P^{\rm cal}(\upsilon',\upsilon;J;B_\alpha A) = P^{\rm obs}(\upsilon',\upsilon;J;B_\alpha A) - P^{\rm cal}(\upsilon',\upsilon;J;B_\alpha A) \quad (30)$$

and $P^{\rm obs}(\upsilon',\ \upsilon;\ J;\ B_\alpha A)$ is the accurate observed line position.

Similarly, for the R-branch transition,

$$R^{\rm IEC}(\upsilon',\upsilon;J;B_\alpha A) = R^{\rm cal}(\upsilon',\upsilon;J;B_\beta A) + \delta R^{\rm cal}(\upsilon',\upsilon;J;B_\alpha A). \quad (31)$$

3. Application and discussion

Equation (27) is used to study the accurate spectra for the ground state of ^{6}LiH and ^{6}LiD molecules. The accurate spectral data of ^{7}LiH and ^{7}LiD, and their accurate effective potentials in which the J-dependent BO breakdown corrections $q^{A,B}(R)$ are not included, are used in producing the IEC data for ^{6}LiH and ^{6}LiD respectively.

An analytical Morse/Lennard–Jones (MLJ) [16] potential and a Rydberg–Klein–Rees (RKR) [21–23] numerical potential of ^{7}LiH are used as the effective potentials $U^{\rm eff}(R)$ to produce the $E^{\rm cal}(v,J)$ for both ^{6}LiH and ^{7}LiH. The RKR potential of ^{7}LiD is used to produce the $E^{\rm cal}(v,J)$ for both ^{6}LiD and ^{7}LiD.

The MLJ potential of ^{7}LiH ground state takes [16]

$$U^{\rm MLJ}(R) = D_e\left\{1 - \left(\frac{R_e}{R}\right)^n \exp[-\phi(R)\cdot z]\right\}^2 \quad (32)$$

where $z=2(R-R_e)/(R+R_e)$, R_e is the equilibrium internuclear distance, D_e is the equilibrium dissociation energy, $n=6$, $\phi(R)$ is a polynomial function of z and R. All the parameters for $U^{\rm MLJ}(R)$ are taken from table 2 of [16]. The numerical RKR potentials $U^{\rm RKR}(R)$ for ^{7}LiH and ^{7}LiD are obtained using the *RKR1* [24] program of Le Roy and the mass-dependent Dunham constants $Y_{l,m}$'s at $m=0$ and 1 given by Dulick *et al.* [8] for ^{7}LiH and ^{7}LiD, respectively. All the potentials are plotted in figure 1(a). The differences between the two ^{7}LiH potentials are shown in figure 1(b).

The calculated energies $E^{\rm cal}(\upsilon,J)$ from the effective potentials $U^{\rm eff}(R)$ are obtained by solving the corresponding radial Schrödinger equation (33) using Le Roy's program *LEVEL7.5* [25] with the eigenvalue convergence criteria of $\mathrm{EPS}=1.0\mathrm{E}-6\,\mathrm{cm}^{-1}$,

$$\left[-\frac{\hbar^2}{2\mu}\frac{d^2}{dR^2} + U^{\rm eff}(R) + \frac{J(J+1)}{2\mu R^2}\hbar\right]^2 \Psi = E^{\rm cal}(\upsilon,J)\Psi. \quad (33)$$

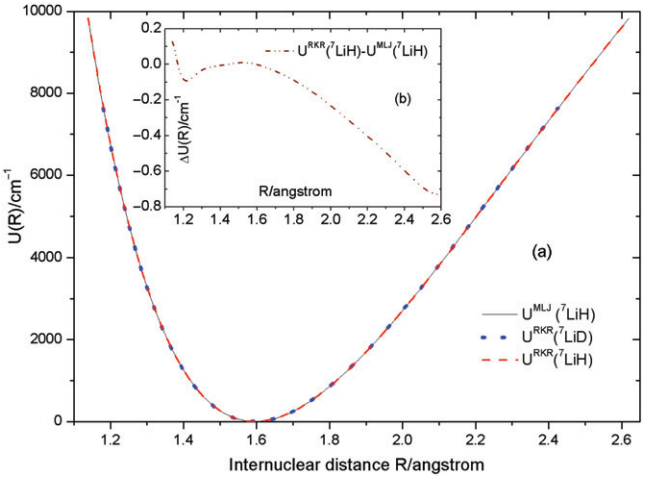


Figure 1. (a) Ground-state potentials of ^{7}LiH and ^{7}LiD; and (b) the difference between two ^{7}LiH potentials. The $U^{MLJ}(R)$ is the Morse/Lennard-Jones (MLJ) rotationless effective potential of ^{7}LiH, which is taken from Coxon and Dickinson [16]. The RKR potentials $U^{RKR}(R)$ for ^{7}LiH and ^{7}LiD are produced from their mass-dependent Dunham constants $Y_{l,m}$ ($m=0$, 1) from Dulick *et al.* [8].

The fundamental constants and atomic masses [26, 27] embedded in the program are adopted. The Schrödinger equations for ^{6,7}LiH are numerically integrated in the range $0.3<R<8.0$ Å with a grid spacing of 0.00175 Å [16] for the MLJ potential. For the numerical RKR potentials the grid spacing is 0.0002 Å for ^{6,7}Li^{1,2}H isotopomers. The convergences are tested for all calculations.

The IEC data obtained using equations (27) and (28) are compared with their accurate experimental energies $E^{exp}(\upsilon, J)$ and the observed line positions for the P and the R branches. The positions of the observed lines are directly taken from Dulick *et al.* [8], while the 'experimental' energies $E^{exp}(\upsilon, J)$ in the range of the observed rovibrational levels are obtained using equation (1) and the mass-dependent Dunham constants $Y_{l,m}$ from [8]. These constants were determined by fitting the observed lines of each isotopomer to equation (1). The uncertainties of most of the observed data used here are about 0.0005 cm^{-1}. The data with large uncertainties ($\delta>0.0010$ cm^{-1}), which are estimated from $\delta=$ observed–calculated values in [8], are not used except those are indicated with '*'. The SCIR formula equation (2) and the $Y_{l,m}$ used to produce the $E^{exp}(\upsilon, J)$ for ^{7}LiH and ^{7}LiD are used to obtain the semi-classical results for isotopic molecules ^{6}LiH and ^{6}LiD respectively. These results are marked with 'SCIR' to compare with the IEC results. The IEC spectral data from the RKR potentials are presented in tables 1–3. The newly predicted IEC data are printed in bold.

Table 1 presents the detailed IEC data and the SCIR results for the P branch of (1,0) band of ^{6}LiH. The observed data of ^{7}LiH used in table 1 have uncertainties $\delta\leq 0.0006$ cm^{-1} [8]. The observed data for ^{6}LiH with large δ (>0.0010 cm^{-1}) are marked with '*' for comparison. The average of the differences between the observed data and all the theoretically obtained data, which are calculated using $\delta_{av}=\Sigma|P(J)_{obs}-P(J)_{cal}|/N_J$, where N_J is the number of the levels under consideration, are also presented in table 1. For the directly calculated data from the $U^{RKR}(R)$ of ^{7}LiH, the δ_{av} are 0.0392 and 0.0385 cm^{-1} for ^{7}LiH and ^{6}LiH respectively. For the SCIR and the IEC results of ^{6}LiH the δ_{av} are 0.0019 and 0.0008 cm^{-1} respectively. From table 1 it is clear that the IEC data are more accurate than the SCIR results. It is also found that for the experimental ^{6}LiH data with large δ, the IEC approach gives either smaller δ than the larger SCIR δ (e.g. $J=17$) or bigger δ than the very small SCIR δ (e.g. $J=1, 2$). For these IEC data the δ cannot indicate the accuracies of the IEC data since the corresponding experimental data of ^{6}LiH are not accurate enough. This is why the IEC data of ^{6}LiH(D) should be produced by using the accurate observed data of ^{7}LiH(D), so that the δ values between the accurate observed/experimental data and the theoretically calculated data can be used to indicate the accuracy of the latter.

Table 2 presents the IEC data for the P branch of (2,1)–(4,3) bands and the R branch of (1,0)–(4,3) bands of ^{6}LiH. The δ_{av}^{cal} of the directly calculated transition positions for either ^{6}LiH or ^{7}LiH is bigger than 0.040 cm^{-1}, and the δ_{av}^{SCIR} of the SCIR data is not less than 0.0020 cm^{-1}, whereas the δ_{av}^{IEC} of the IEC results is less than 0.0010 cm^{-1} for each band in table 2.

Table 3 presents the IEC data for the P and the R branches of (1,0)–(5,4) bands of ^{6}LiD. For each band in table 3, the δ_{av}^{cal} of the directly calculated transition positions for either ^{6}LiD or ^{7}LiD is bigger than 0.016 cm^{-1}, and the δ_{av}^{SCIR} of the SCIR data is not less than 0.0016 cm^{-1}, whereas the δ_{av}^{IEC} of the IEC results is less than 0.0010 cm^{-1}. The comparison between the IEC data from different potentials and the SCIR results are presented in figures 2–7.

All the theoretical energies $E^{cal}(\upsilon,J)$ are shifted such that $E^{exp}(\upsilon=0, J=0)=E^{cal}(\upsilon=0, J=0)$, and the energy errors $\delta E(\upsilon,J)=E^{exp}(\upsilon,J)-E^{cal}(\upsilon,J)$ are compared with each other in figures 2 and 3 for ^{6,7}LiH and in figure 4 for ^{6,7}LiD. In these figures, the IEC energies of ^{6}Li^{1,2}H

Table 1. Comparisons between the P branch data of the (1,0) band (in cm^{-1}) for the $X^1\Sigma^+$ state of ^{6,7}LiH isotopomers.[#]

^{7}LiH			^{6}LiH					
P(J)	cal	obs/Ref. [8]	cal	obs/Ref. [8]	IEC	obs-IEC	SCIR	obs-SCIR
1	1344.8567	1344.9005	1358.2083	1358.2525*	1358.2521	0.0004*	1358.2525	0.0000*
2	1329.6437	1329.6872	1342.6757	1342.7200*	1342.7192	0.0008*	1342.7199	0.0001*
3	1314.0470	1314.0902	1326.7479	1326.7896	1326.7911	−0.0015	1326.7919	−0.0023
4	1298.0876	1298.1306	1310.4466	1310.4886	1310.4896	−0.0010	1310.4905	−0.0019
5	1281.7865	1281.8291	1293.7937	1293.8356	1293.8363	−0.0007	1293.8375	−0.0019
6	1265.1648	1265.2071	1276.8116	1276.8535	1276.8539	−0.0004	1276.8550	−0.0015
7	1248.2437	1248.2857	1259.5219	1259.5632	1259.5639	−0.0007	1259.5650	−0.0018
8	1231.0441	1231.0857	1241.9467	1241.9875	1241.9883	−0.0008	1241.9895	−0.0020
9	1213.5873	1213.6284	1224.1078	1224.1481	1224.1489	−0.0008	1224.1502	−0.0021
10	1195.8936	1195.9342	1206.0267	1206.0670	1206.0673	−0.0003	1206.0686	−0.0016
11	1177.9837	1178.0236	1187.7247	1187.7641	1187.7646	−0.0005	1187.7661	−0.0020
12	1159.8776	1159.9171	1169.2229	1169.2612	1169.2624	−0.0012	1169.2636	−0.0024
13	1141.5952	1141.6337	1150.5416	1150.5794	1150.5801	−0.0007	1150.5815	−0.0021
14	1123.1557	1123.1935	1131.7011	1131.7380	1131.7389	−0.0009	1131.7401	−0.0021
15	1104.5781	1104.6148	1112.7208	1112.7571	1112.7575	−0.0004	1112.7589	−0.0018
16	1085.8808	1085.9165	1093.6200	1093.6552	1093.6557	−0.0005	1093.6569	−0.0017
17	1067.0815	1067.1161	1074.4170	1074.4493*	1074.4516	−0.0023*	1074.4528	−0.0035*
18	1048.1976	1048.2310	1055.1298	1055.1624	1055.1632	−0.0008	1055.1644	−0.0020
19	1029.2458	1029.2782	1035.7758	1035.8072	1035.8082	−0.0010	1035.8090	−0.0018
20	1010.2422	1010.2732	1016.3716	1016.4017	1016.4026	−0.0009	1016.4033	−0.0016
21	991.2022	991.2319	996.9330	–	996.9627	–	996.9633	–
22	972.1408	972.1691	977.4758	–	977.5041	–	977.5045	–
23	953.0723	953.0989	958.0141	–	958.0407	–	958.0415	–
24	934.0100	934.0354	938.5623	–	938.5877	–	938.5883	–
25	914.9670	914.9913	919.1335	–	919.1578	–	919.1582	–
26	895.9555	895.9782	899.7403	–	899.7630	–	899.7638	–
δ_{av}[‡]	0.0392	–	0.0385	–	–	0.0008	–	0.0019

#'cal' denotes the calculated results for both ^{7}LiH and ^{6}LiH from the ^{7}LiH RKR potential; 'obs' denotes the observed data ($\delta \le 0.0006\,cm^{-1}$) from Dulick *et al.* [8]; 'IEC' denotes the IEC data obtained from $P(J,{}^6LiH)_{IEC} = P(J,{}^6LiH)_{cal} + [P(J,{}^7LiH)_{obs} - P(J,{}^7LiH)_{cal}]$; 'SCIR' denotes the data obtained from equation (2) and all the mass-dependent Dunham constants of ^{7}LiH given by [8].

*The data have large experimental δ. The δ of the observed ^{6}LiH spectra are 0.0020, 0.0023, and $-0.0013\,cm^{-1}$ for $J = 1$, 2, and 17, respectively [8].

‡$\delta_{av} = \Sigma|P(J)_{obs} - P(J)_{cal}|/N_J$. N_J is the number of levels in consideration. The data with large experimental $\delta > 0.0010\,cm^{-1}$ are not included.

are obtained by using either $U^{MLJ}(R)$ or $U^{RKR}(R)$ potentials of $^7Li^{1,2}H$.

From figures 2–4 it can be seen clearly that the IEC errors of $^6Li^{1,2}H$ (solid line —) are much smaller than the errors of the SCIR results (dashed line - - -), as predicted by equation (23). This shows that the IEC energies of $^6Li^{1,2}H$ are more accurate than the corresponding SCIR data. The IEC error curve shown in figure 2 is also plotted in figure 3 (with dash dotted line - - -) for comparison. From figure 3 it can be seen that the ^{6}LiH IEC errors (—) from the ^{7}LiH $U^{RKR}(R)$ potential are slightly different to those (- - -) from the $U^{MLJ}(R)$ of ^{7}LiH. For most of the levels the former presents smaller errors than the latter. The comparison between the two IEC errors shows that the accuracies of the IEC energies are influenced by the effective potentials.

The results for ^{6}LiD IEC errors shown in figure 4 satisfy equation (23) very well, i.e. it can be seen clearly that $0 < \delta E^{IEC}(^6LiD)/\delta E^{SCIR}(^6LiD)$ and $\delta E^{IEC}(^6LiD)/\delta E^{SCIR}(^6LiD) < 1$. The results shown in figures 2 and 3 demonstrate that most of the data satisfy equation (23) well, except for some of the high-J level data, e.g. $J = 20$ of $v = 0$ in figures 2(a) and 3(a), for which

Table 2. The P and R branch IEC vibration–rotational line positions (in cm^{-1}) for the $X^1\Sigma^+$ state of ^6LiH. The newly obtained IEC data are presented in bold.[#]

J	P(2,1)	P(3,2)	P(4,3)	R(1,0)	R(2,1)	R(3,2)	R(4,3)
0	–	–	–	**1388.0450**	1341.8357	–	–
1	1312.9101	**1268.5042**	–	1402.2644	1355.6325	**1309.9575**	**1265.1559**
2	1297.8235	**1253.8540**	**1210.7299**	1416.0032	1368.9579	**1322.8755**	**1277.6727**
3	1282.3482	**1238.8219**	**1196.1340**	1429.2435	1381.7943	1335.3129	**1289.7173**
4	1266.5059	**1223.4301**	**1181.1812**	1441.9655	1394.1211	1347.2498	**1301.2676**
5	1250.3182	**1207.6980**	**1165.8957**	1454.1501	1405.9205	**1358.6678**	–
6	1233.8070	1191.6475	**1150.2965**	1465.7807	1417.1756	1369.5521	**1322.8229**
7	1216.9935	1175.2994	**1134.4035**	1476.8397	1427.8698	1379.8840	**1332.7951**
8	1199.8996	1158.6763	**1118.2385**	1487.3116	1437.9873	1389.6498	–
9	1182.5465	1141.7970	**1101.8215**	1497.1810	1447.5131	**1398.8340**	–
10	1164.9551	1124.6829	**1085.1729**	1506.4342	1456.4346	1407.4244	**1359.3110**
11	1147.1461	1107.3555	**1068.3127**	1515.0585	1464.7375	**1415.4073**	–
12	1129.1407	1089.8331	**1051.2594**	1523.0416	1472.4117	**1422.7711**	–
13	1110.9580	1072.1352	**1034.0336**	1530.3733	**1479.4457**	**1429.5062**	–
14	1092.6186	1054.2831	**1016.6535**	**1537.0439**	**1485.8309**	**1435.6052**	**1386.2621**
15	1074.1409	1036.2948	**999.1362**	**1543.0445**	**1491.5584**	–	–
16	1055.5439	–	**981.5018**	**1548.3686**	**1496.6227**	**1445.8544**	–
17	1036.8454	999.9775	**963.7633**	**1553.0106**	–	**1449.9923**	–
18	–	**981.6840**	**945.9412**	**1556.9650**	–	**1453.4676**	–
19	999.2136	**963.3235**	**928.0503**	–	–	–	–
20	980.3133	**944.9101**	**910.1038**	–	–	–	–
21	**961.3774**	**926.4588**	**892.1175**	–	–	–	–
22	**942.4209**	**907.9853**	–	–	–	–	–
23	**923.4569**	**889.5031**	–	–	–	–	–
24	**904.5032**	–	–	–	–	–	–
26	**866.6633**	–	–	–	–	–	–
$\delta_{av}^{cal}(^6LiH)^{\ddagger}$	0.0404	0.0416	–	0.0444	0.0447	–	–
$\delta_{av}^{cal}(^7LiH)^{\ddagger}$	0.0410	0.0421	0.0429	0.0451	0.0456	0.0502	0.0483
$\delta_{av}^{IEC}(^6LiH)^{\ddagger}$	0.0008	0.0007	–	0.0008	0.0009	–	–
$\delta_{av}^{SCIR}(^6LiH)^{\ddagger}$	0.0020	0.0020	–	0.0023	0.0023	–	–

#All the IEC data are obtained using $P(J,^6LiH)_{IEC} = P(J,^6LiH)_{cal} + [P(J,^7LiH)_{obs} - P(J,^7LiH)_{cal}]$ and $R(J,^6LiH)_{IEC} = R(J,^6LiH)_{cal} + [R(J,^7LiH)_{obs} - R(J,^7LiH)_{cal}]$. The calculated data are obtained from the ^7LiH RKR potential; the observed data are from Dulick *et al.* [8] with $\delta \le 0.0010\,cm^{-1}$.
‡Symbols have similar meanings as the δ_{av} in table 1. All the δ_{av} in the same column are for the same levels with experimental $\delta \le 0.0010\,cm^{-1}$.

$\delta E^{IEC}(^6LiH)/\delta E^{SCIR}(^6LiH) < 0$. One reason for this case is the influence of the accidental errors of the 'experimental data' $E^{exp}(^6LiH)$, which is not considered in equation (23), since the IEC errors and the accidental errors are of the same order (which is estimated to be $\sim 0.001\,cm^{-1}$, i.e. the largest uncertainty of the measured line positions), it may influence the sign of the IEC errors sometimes. But even so, it still gives $|\delta E^{IEC}(^6LiH)/\delta E^{SCIR}(^6LiH)| < 1$, i.e. smaller IEC errors than SCIR errors.

The difference between the theoretical value and the experimental value includes the sum of the systematic error and the accidental error. The accidental errors introduced in the numerical computations, which are in the order of the convergence criteria of the eigenvalues ($\sim 10^{-5}\,cm^{-1}$), can be neglected compared with the total errors. The uncertainties (the observed–calculated values in [16]) of the observed line positions, which can be taken to be the accidental errors of the experimental values $E^{exp}(v,J)$, are also small (less than $0.001\,cm^{-1}$), and can be neglected in the large total errors. Therefore, all the large errors (e.g. the high-J levels) shown in figures 2–4 can be assumed as systematic errors, since the accidental errors are so small that they may be neglected compared with the total errors.

Table 3. The P and R branch IEC vibration–rotational line positions (in cm^{-1}) for the $X^1\Sigma^+$ state of ^{6}LiD. The newly obtained IEC data are presented in bold #.

J	P(1,0)	P(2,1)	P(3,2)	P(4,3)	P(5,4)	R(1,0)	R(2,1)	R(3,2)	R(4,3)	R(5,4)
0	–	–	–	–	–	–	–	**1003.2497**	–	–
1	**1038.8221**	**1012.6142**	–	**961.4339**	–	**1064.2931**	**1037.5205**	**1011.1757**	–	–
2	**1029.9562**	**1003.9395**	**978.3415**	**953.1381**	–	**1072.3834**	**1045.4294**	**1018.9039**	**992.7861**	–
3	1020.9133	995.0927	**969.6873**	**944.6735**	**920.0218**	1080.2640	**1053.1300**	**1026.4283**	**1000.1345**	–
4	1011.7012	986.0786	**960.8674**	**936.0463**	**911.5818**	1087.9266	**1060.6168**	**1033.7405**	**1007.2741**	–
5	1002.3264	976.9042	**951.8899**	**927.2626**	**902.9915**	1095.3660	**1067.8826**	**1040.8355**	–	**987.9434**
6	992.7976	967.5770	942.7626	**918.3300**	**894.2509**	1102.5753	**1074.9216**	**1047.7059**	**1020.9027**	–
7	983.1207	958.1045	933.4913	**909.2554**	–	1109.5491	**1081.7288**	**1054.3477**	**1027.3805**	**1000.7964**
8	973.3034	948.4932	924.0820	**900.0460**	–	1116.2814	**1088.2976**	**1060.7540**	**1033.6269**	**1006.8816**
9	963.3530	938.7507	914.5443	**890.7078**	**867.2161**	–	**1094.6225**	**1066.9214**	**1039.6357**	–
10	953.2767	928.8840	904.8826	**881.2495**	**857.9548**	1128.9990	**1100.6985**	**1072.8418**	–	–
11	943.0816	918.9002	895.1064	**871.6762**	**848.5822**	1134.9738	**1106.5212**	**1078.5117**	–	–
12	932.7750	908.8060	885.2214	**861.9961**	**839.1013**	1140.6871	**1112.0838**	–	**1056.1869**	–
13	922.3638	898.6090	875.2334	**852.2154**	**829.5211**	1146.1321	**1117.3842**	**1089.0815**	**1061.1969**	–
14	911.8551	888.3152	865.1514	**842.3389**	**819.8477**	**1151.3067**	–	–	–	–
15	901.2561	**877.9326**	854.9808	**832.3760**	**810.0869**	**1156.2053**	**1127.1763**	**1098.5954**	–	–
16	890.5727	867.4671	844.7282	**822.3317**	**800.2466**	**1160.8254**	**1131.6636**	–	–	–
17	879.8129	856.9253	834.4001	**812.2130**	**790.3320**	**1165.1613**	**1135.8693**	**1107.0245**	–	–
18	868.9825	846.3142	824.0034	**802.0258**	–	**1169.2115**	**1139.7939**	**1110.8222**	–	–
19	858.0882	835.6399	813.5445	**791.7778**	–	**1172.9748**	–	**1114.3407**	–	–
20	847.1365	824.9083	803.0276	**781.4709**	–	–	–	–	–	–
21	836.1331	814.1265	792.4629	–	–	**1179.6193**	–	–	–	–
22	825.0850	803.2991	–	–	–	–	–	–	–	–
23	–	792.4329	–	–	–	**1185.0826**	–	–	–	–
24	802.8764	781.5335	–	–	–	**1187.3665**	–	–	–	–
25	791.7277	–	–	–	–	–	–	–	–	–
26	**780.5566**	–	–	–	–	–	–	–	–	–
$\delta_{av}^{cal}(^6LiD)^{\ddagger}$	0.0166	0.0160	0.0166	–	–	0.0177	–	–	–	–
$\delta_{av}^{cal}(^7LiD)^{\ddagger}$	0.0173	0.0168	0.0172	0.0192	0.0272	0.0184	0.0184	0.0194	0.0203	0.0249
$\delta_{av}^{IEC}(^6LiD)^{\ddagger}$	0.0008	0.0009	0.0006	–	–	0.0006	–	–	–	–
$\delta_{av}^{SCIR}(^6LiD)^{\ddagger}$	0.0017	0.0016	0.0016	–	–	0.0016	–	–	–	–

#All the IEC data are obtained using $P(J,^6LiD)_{IEC} = P(J,^6LiD)_{cal} + [P(J,^7LiD)_{obs} - P(J,^7LiD)_{cal}]$ and $R(J,^6LiD)_{IEC} = R(J,^6LiD)_{cal} + [R(J,^7LiD)_{obs} - R(J,^7LiD)_{cal}]$. All the calculated data are obtained from the ^{7}LiD RKR potential; all the observed data are from [8] with $\delta \leq 0.0010\,cm^{-1}$.
‡Symbols have similar meanings to those in table 2.

From the IEC errors (solid lines —) shown in figures 2–4, it is clear that the small IEC systematic errors (which are the difference between the systematic errors of the two isotopic molecules, e.g. $\delta E^{IEC}(^6LiH) = \delta E^{cal}(^6LiH) - \delta E^{cal}(^7LiH)$) take the place of the large systematic errors (e.g. $\delta E^{cal}(^6LiH)$) introduced from the effective potentials $U^{eff}(R)$, in which the J-dependent BO breakdown corrections $q^{A,B}(R)$ should be included. The IEC results (—) for high J levels from the $U^{MLJ}(R)$ (in figure 2) are good examples to illustrate the special significance of the IEC approach.

From figure 2 it is observed that the $U^{MLJ}(R)$ is an accurate rotationless effective potential for ^{7}LiH ground state, for which the eigenvalues for lower J of all υ states under consideration are very accurate ($\delta \approx 0$), while for higher J states (e.g. $J > 5$ for $\upsilon = 0$ in figure 2(a) and $J > 10$ for other υ states in figure 2(b)–(d), the errors introduced from the J-dependent BO breakdown effects are greatly increased. For lower J states of ^{6}LiH the ^{7}LiH $U^{MLJ}(R)$ still gives accurate results (though not as accurate as that of ^{7}LiH), and it still gives smaller errors (line with filled circle –●–) than the SCIR (- - -). For higher J states, like that of ^{7}LiH, the errors increase so

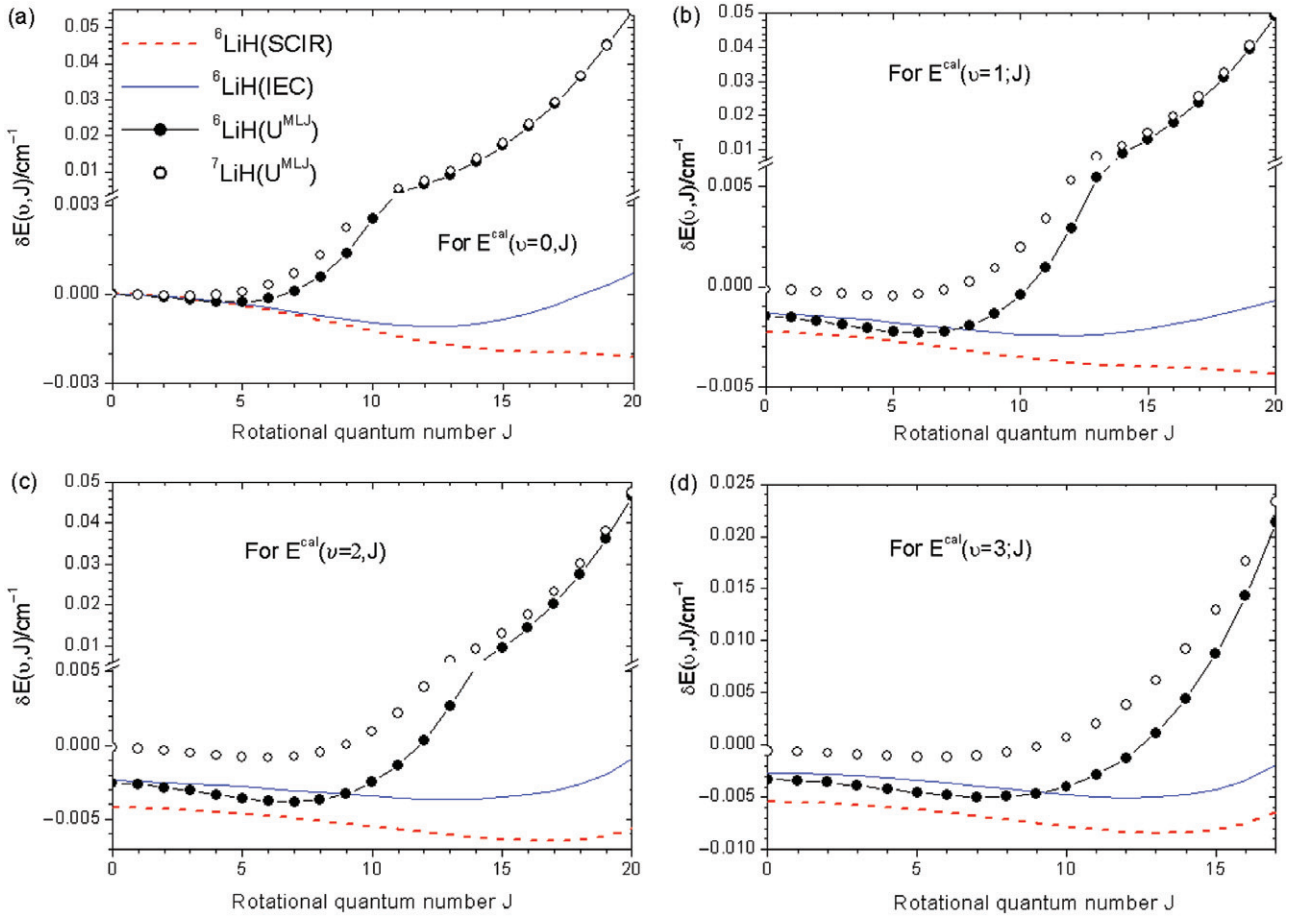


Figure 2. Errors $\delta E(\upsilon,J) = E^{\exp}(\upsilon,J) - E^{\mathrm{cal}}(\upsilon,J)$ for the calculated vibration–rotational level energies $E^{\mathrm{cal}}(\upsilon,J)$ for the $X^1\Sigma^+$ state of ^{6,7}LiH. $E^{\exp}(\upsilon,J)$ are obtained using equation (1) and the experimental Dunham constants $Y_{l,m}$ from Dulick *et al.* [8]. All the $E^{\mathrm{cal}}(0,0)$ are shifted to $E^{\mathrm{cal}}(0,0) = E^{\exp}(0,0)$. Dashed lines (- - -) for ^{6}LiH SCIR levels obtained from isotope relation equation (2) and the ^{7}LiH $Y_{l,m}$ from [8]; circles (○) for ^{7}LiH levels obtained from $U^{\mathrm{MLJ}}(R)$; filled circles connected by solid lines (–●–) for ^{6}LiH levels obtained from ^{7}LiH $U^{\mathrm{MLJ}}(R)$; solid lines (—) for ^{6}LiH IEC levels from $U^{\mathrm{MLJ}}(R)$, $\delta E^{\mathrm{IEC}}(^6\mathrm{LiH}) = \delta E^{\mathrm{cal}}(^6\mathrm{LiH}) - \delta E^{\mathrm{cal}}(^7\mathrm{LiH})$.

much that they (circle ○ for ^{7}LiH and –●– for ^{6}LiH) are far larger than the SCIR errors (- - -). For the ^{6}LiH IEC results (—) at the same potential, the errors are stable and far smaller than the SCIR errors (- - -) and the $\delta E^{\mathrm{cal}}(^6\mathrm{LiH})$ (–●–). The physical factors are the accurate isotope effects, which make $\delta E^{\mathrm{cal}}(^6\mathrm{LiH}) \approx \delta E^{\mathrm{cal}}(^7\mathrm{LiH})$ for higher J states, so that most of the J-dependent errors of ^{6}LiH can be eliminated relative to those of ^{7}LiH. These results show that the IEC approach is an effective way to eliminate large systematic errors. Similar conclusions can also be drawn from the errors of the corresponding IEC transition positions, which are shown in figures 5 and 6.

All the calculated transition positions, which are the differences between all two-level energies and are not influenced by their original eigenvalues, are compared with reference to their errors with respect to the accurately observed values from [8]. The errors for the P and the R branches of ($\upsilon' = 1 - \upsilon = 0$), for short (1,0) transition bands and the P branches of (2,1) and (3,2) bands, for which the accurately observed data are available for comparison, are shown in figure 5 for ^{6,7}LiH and figure 6 for ^{6,7}LiD ground states. The IEC errors are calculated using $\delta P^{\mathrm{IEC}}(^6\mathrm{Li}^{1,2}\mathrm{H}) = P^{\mathrm{obs}}(^6\mathrm{Li}^{1,2}\mathrm{H}) - P^{\mathrm{IEC}}(^6\mathrm{Li}^{1,2}\mathrm{H})$ or $\delta P^{\mathrm{IEC}}(^6\mathrm{Li}^{1,2}\mathrm{H}) = \delta P^{\mathrm{cal}}(^6\mathrm{Li}^{1,2}\mathrm{H}) - \delta P^{\mathrm{IEC}}(^7\mathrm{Li}^{1,2}\mathrm{H})$ for the P branches and similar formulae for the R branches.

From figure 5 it is clear that the ^{6}LiH IEC spectra from either $U^{\mathrm{MLJ}}(R)$ (dash-dotted line - - -) or $U^{\mathrm{RKR}}(R)$ (—) of ^{7}LiH are more accurate than the SCIR

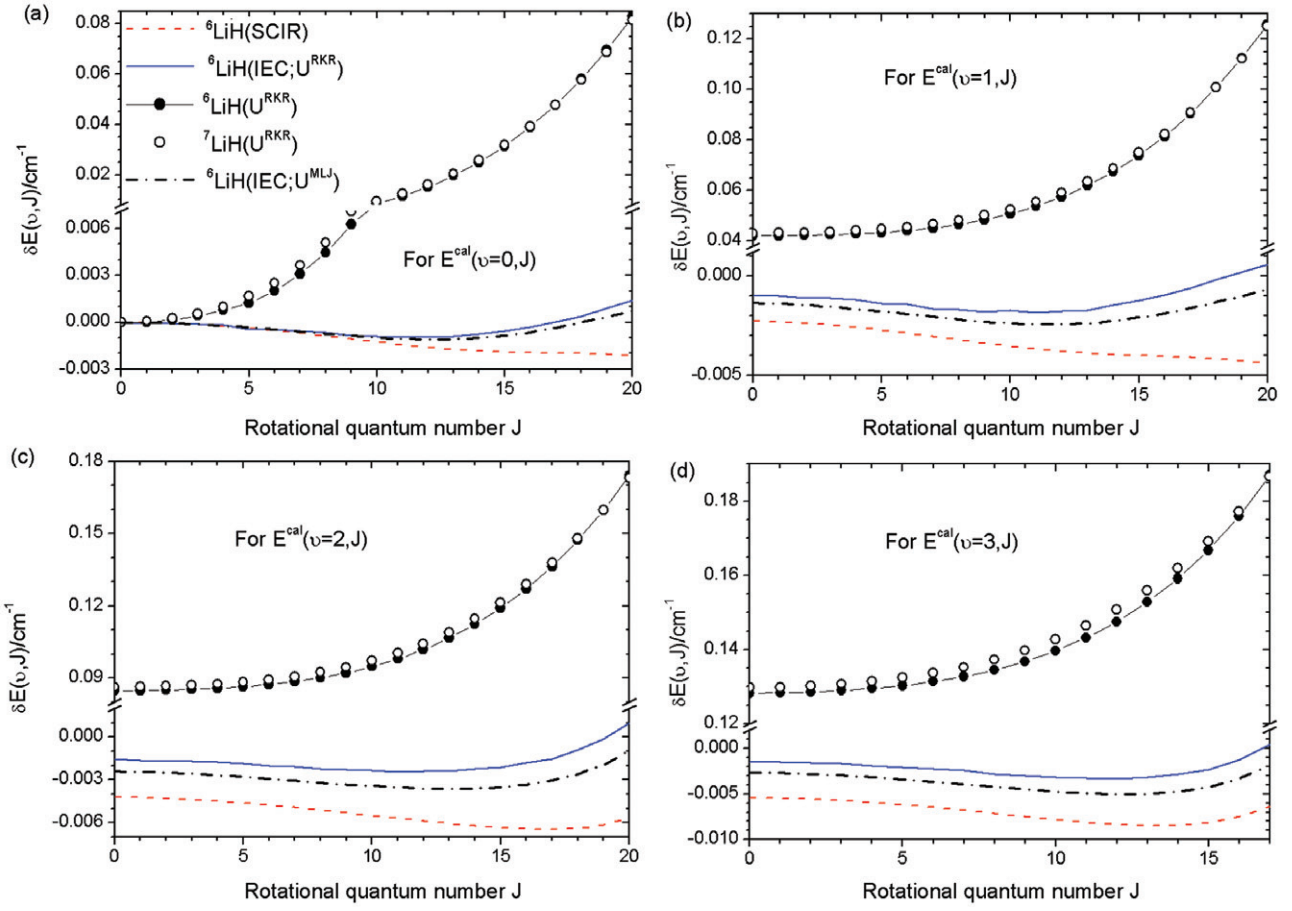


Figure 3. Same as figure 2, except that the calculated vibration–rotational level energies $E^{\mathrm{cal}}(\upsilon,J)$ are from the $U^{\mathrm{RKR}}(R)$ of ^{7}LiH and dash-dotted lines (- -) are taken from figure 2 for ^{6}LiH IEC spectra from $U^{\mathrm{MLJ}}(R)$.

spectra (- - -). Comparing the two IEC results for ^{6}LiH, those from U^{RKR}(R) are slightly better than that from the U^{MLJ}(R) of ^{7}LiH. The predicted IEC results from U^{RKR}(R) are presented in tables 1 and 2 for ^{6}LiH.

The errors shown in figure 5 indicate that for ^{6}LiH, the IEC spectra at lower J states are similar to or slightly better than the directly obtained spectra from the $U^{\mathrm{MLJ}}(R)$ (–▲–) or the $U^{\mathrm{RKR}}(R)$ (–●–) of ^{7}LiH, while at higher J levels they are far more accurate than the directly obtained data from the same effective potentials.

Similarly, figure 6 shows that the ^{6}LiD IEC spectra (—) from the U^{RKR}(R) of ^{7}LiD are more accurate than the corresponding SCIR spectra (- - -), and are far more accurate than the directly calculated results (–●–) from U^{RKR}(R) of ^{7}LiD for all levels.

For most of the ^{6}LiH(D) IEC spectra from the RKR potentials of ^{7}LiH(D), it is found from figures 5 and 6 that the IEC errors fall within the range 0–0.0010 cm^{-1}. The accuracies of such IEC data are then conservatively estimated to be $0.0010 + u_{\mathrm{obs}}$ cm^{-1}, where u_{obs} are the uncertainties of the ^{7}LiH(D) observed line positions. For most of the observed ^{7}LiH(D) data, the uncertainties u_{obs} are in the range 0.0005–0.0010 cm^{-1} [8], so the accuracy of the corresponding new IEC spectra for ^{6}LiH(D) are estimated to be $0.0015 \sim 0.0020$ cm^{-1}. The IEC data may also be improved by reducing the systematic errors introduced by the IEC approach, if necessary, since such systematic errors are almost stable in general and can be estimated using a few accurate observed data instances at the target molecular state.

The predicted IEC results in table 3 and the SCIR data for the R branches of (1,0) and (2,1) bands of ^{6}LiD are compared in figure 7 with other observed data available from Maki *et al.* [28] and the recent literature

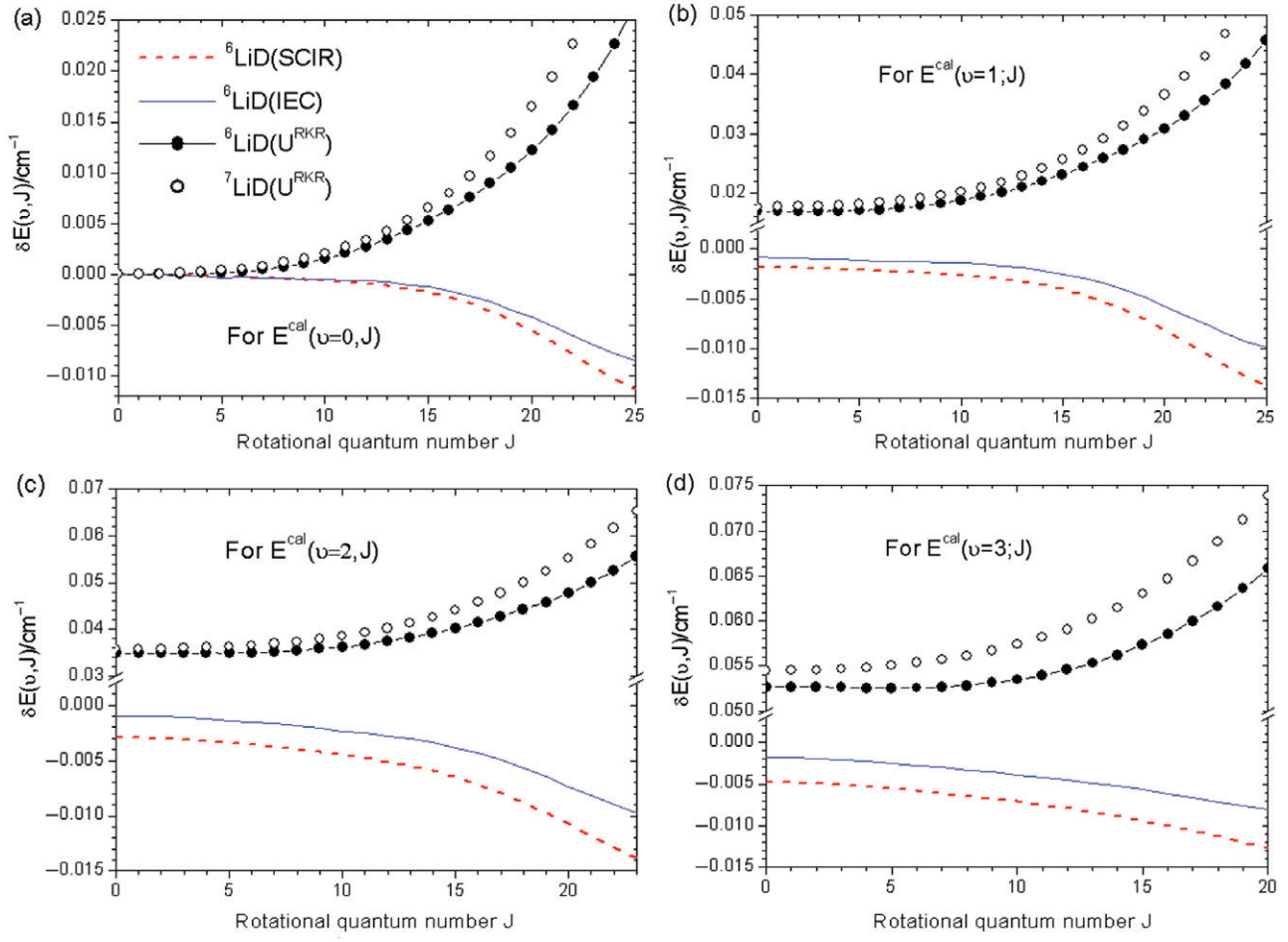


Figure 4. Errors $\delta E(\upsilon,J) = E^{exp}(\upsilon,J) - E^{cal}(\upsilon,J)$ for the calculated vibration–rotational level energies $E^{cal}(\upsilon,J)$ for the $X^1\Sigma^+$ state of ^{6,7}LiD isotopic molecules. $E^{exp}(\upsilon,J)$ are obtained using equation (1) and the experimental Dunham constants $Y_{l,m}$ from Dulick *et al.* [8]. All the $E^{cal}(0,0)$ are shifted so that $E^{cal}(0,0) = E^{exp}(0,0)$. Dashed lines (- - -) for ^{6}LiD SCIR levels obtained from isotope relation equation (2) and the ^{7}LiD $Y_{l,m}$ from [8]. Circles (○) for ^{7}LiD levels obtained from $U^{RKR}(R)$ ($\delta E^{cal}(^7\text{LiD})$); filled circles connected by solid lines (–●–) for ^{6}LiD levels obtained from ^{7}LiD $U^{RKR}(R)$ ($\delta E^{cal}(^6\text{LiD})$); solid lines (—) for ^{6}LiD IEC levels from $U^{RKR}(R)$, $\delta E^{IEC}(^6\text{LiD}) = \delta E^{cal}(^6\text{LiD}) - \delta E^{cal}(^7\text{LiD})$.

direct potential fits (DPF) results from Coxon and Dickinson [16]. In the DPF study the Hamiltonian, which includes a parameterized potential energy function and the BO breakdown functions (including both *J*-dependent and *J*-independent corrections) are obtained from the observed frequencies of [28] and other observed data [16] are used to produce the DPF spectral data. The SCIR data are from the same $Y_{l,m}$ [8] as those in figure 6. The accuracies of the observed data were assessed as as 0.0008 cm^{-1} in the DPF study [16]. It is clear from figure 7 that the accuracy of the IEC (—) data is between that of the DPF (dotted line ···) and the SCIR (- - -) data, and is approximately the average of the two. These results show that the IEC approach is a reliable method of predicting more accurate spectra for isotopic molecules than the SCIR method.

The accurate spectra and the RKR potential of ^{7}LiH are also applied to produce the IEC results of ^{6}LiD and ^{7}LiD, for which the IEC approach still gives more accurate results than those from the $Y_{l,m}$ of ^{7}LiH and the SCIR in equation (2).

All the above IEC results for ^{6}LiH and ^{6}LiD show that the IEC approach can be used to predict accurate isotopic molecular spectra based on good effective potentials. Both the accurate analytic rotationless potential and the numerical RKR

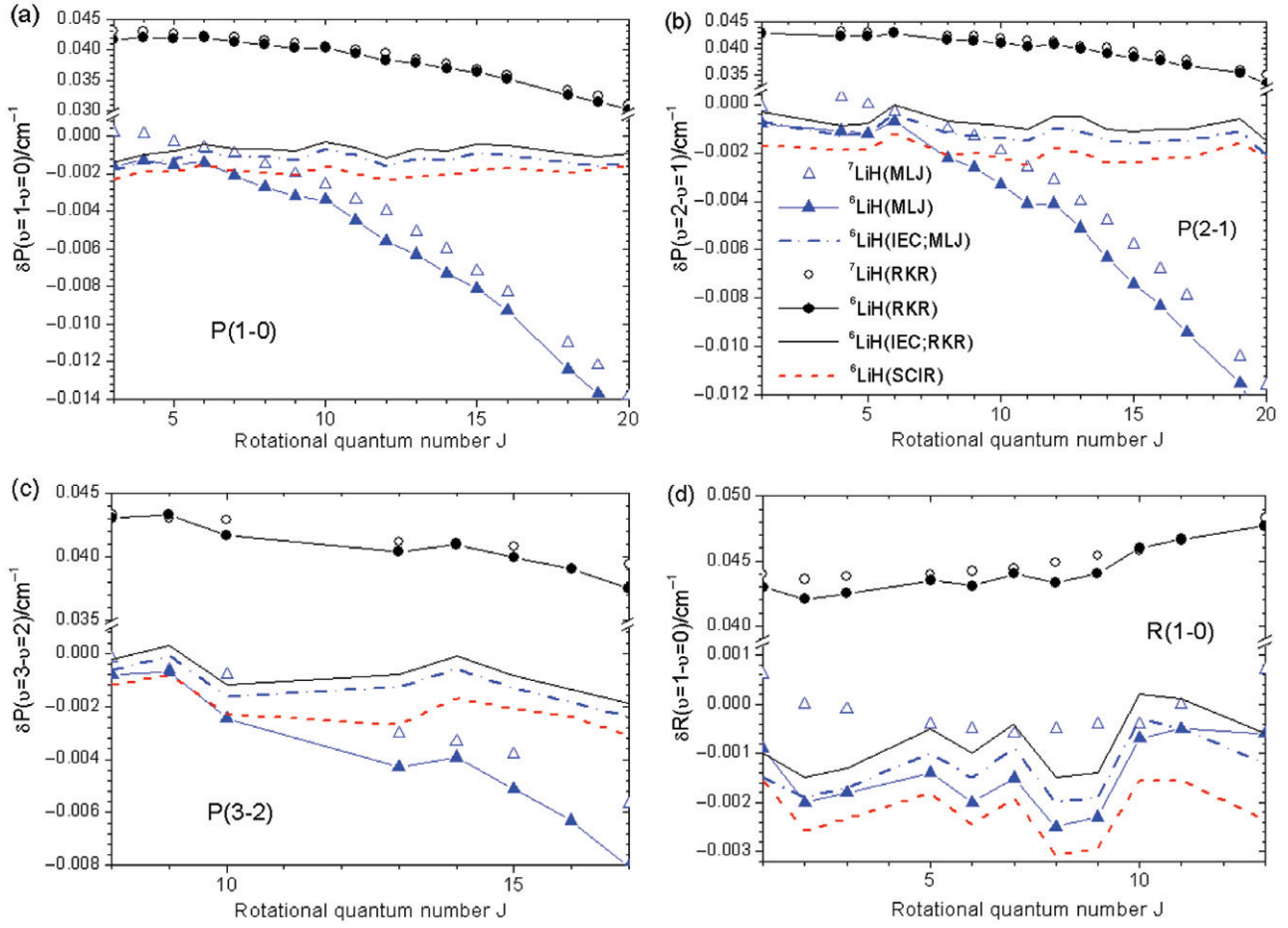


Figure 5. Errors of the calculated line positions for the $X^1\Sigma^+$ state of ^{6,7}LiH isotopic molecules. All the observed data are taken from [8]. Triangles (Δ) for ^{7}LiH spectra obtained from $U^{MLJ}(R)$; filled triangles connected by solid lines (–▲–) for ^{6}LiH spectra obtained from $U^{MLJ}(R)$; dash-dotted lines (-·-·) for ^{6}LiH IEC spectra from $U^{MLJ}(R)$ and the observed ^{7}LiH spectra. Circles (○) for ^{7}LiH spectra obtained from the $U^{RKR}(R)$ of ^{7}LiH; filled circles connected by solid lines (–●–) for ^{6}LiH spectra obtained from ^{7}LiH $U^{RKR}(R)$; solid lines (—) for ^{6}LiH IEC spectra from $U^{RKR}(R)$ and the observed ^{7}LiH spectra; dashed lines (- - - -) for ^{6}LiH SCIR spectra obtained from isotope relation equation (2) and the ^{7}LiH $Y_{l,m}$ from [8].

potential can be used as good effective potential in the IEC approach. The qualities of the effective potentials do influence the qualities of the IEC data.

4. Conclusions

The proposed IEC approach presented here can easily be adopted to study the accurate vibration–rotational spectra of isotopic molecules with reduced masses $\mu(B_\alpha A) \approx \mu(B_\beta A)$. The accuracy of the predicted IEC spectra for $B_\beta A$ is influenced by the accuracy of the experimental data (of $B_\alpha A$) and the effective potential of isotopic molecule $B_\alpha A$. Although some small systematic errors are introduced into the IEC results, the IEC approach is proved to predict more accurate transition frequencies than the semi-classical isotope relation, equation (2). On condition that the spectral data are available for only one isotopomer, the IEC approach can be used to predict more accurate spectral data for other isotopomers. Some new spectral data are obtained for the (1,0)–(4,3) bands of ^{6}LiH in table 2 and for the (1,0)–(5,4) bands of ^{6}LiD in table 3 from the RKR potentials of ^{7}LiH and ^{7}LiD using the IEC method. The IEC approach is also an effective way to eliminate the large systematic errors caused by neglecting the *J*-dependent BO breakdown corrections.

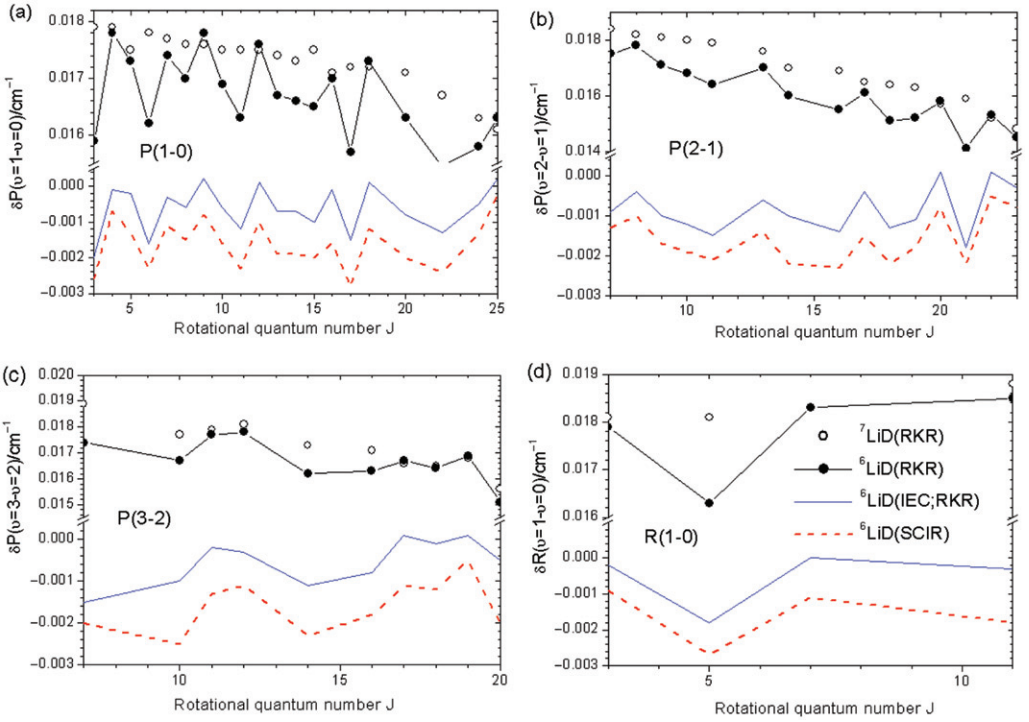


Figure 6. Errors for the calculated line positions for the $X^1\Sigma^+$ state of ^{6,7}LiD isotopic molecules. Circles (○) for ^{7}LiD spectra obtained from the $U^{RKR}(R)$ of ^{7}LiD; filled circles connected by solid lines (–●–) for ^{6}LiD spectra obtained from ^{7}LiD $U^{RKR}(R)$; solid lines (—) for ^{6}LiD IEC spectra from $U^{RKR}(R)$ and the observed ^{7}LiD spectra; dashed lines (- - - -) for ^{6}LiD SCIR spectra obtained from isotope relation equation (2) and the ^{7}LiD $Y_{l,m}$ from [8].

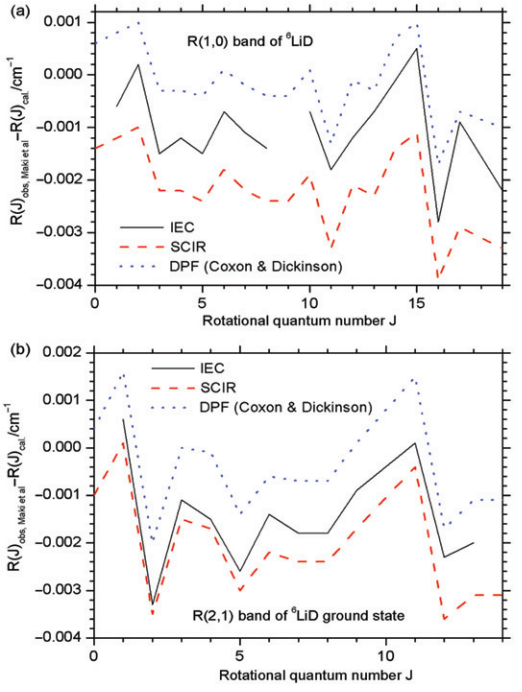


Figure 7. Comparisons between the errors of the predicted IEC spectra and the direct potential fits (DPF) results for the R(1,0) (a) and R(2,1) (b) transition bands of ^{6}LiD. The observed data (obs) are from Maki *et al.* [28], the DPF data are from Coxon and Dickinson [16], the IEC data from table 3 in this work, the SCIR data are from equation (2) and the $Y_{l,m}$ [8] of ^{7}LiD. Dotted lines (. . .) for $\delta R(\mathrm{DPF}) = R(\mathrm{obs}) - R(\mathrm{DPF})$; solid lines (—) for $\delta R(\mathrm{IEC}) = R(\mathrm{obs}) - R(\mathrm{IEC})$; dashed lines (- - - -) for $\delta R(\mathrm{SCIR}) = R(\mathrm{obs}) - R(\mathrm{SCIR})$.

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